

Project Status Report (PSR): SPECTRUM Market Intelligence Dashboard

Project: SPECTRUM Market Intelligence Dashboard **Report Period:** February 5-6, 2026 (Initial Build Sprint)

Prepared By: Cole Prather **Date Issued:** February 6, 2026 **Report Version:** 1.0

Reference Documents: - Project Overview Document (POD): /project-pods/spectrum_dashboard_POD.md
- Data Pipeline Documentation: /market-analytics-dashboard/DATA_PIPELINE.md

1. Executive Summary

SPECTRUM was built from scratch over a 2-day sprint (Feb 5-6, 2026) and is now **fully operational** with real market data. The dashboard monitors 54 stocks and 8 cryptocurrencies across five scoring dimensions (Fundamentals, Technicals, Valuation, Momentum, Analyst) using 29 normalized metrics. Technical indicators (RSI, MACD, Bollinger Bands, SMA alignment) are computed client-side from 5-year price history. All 62 assets have curated intel briefs providing thesis, catalyst, risk, and watchlist context. The data pipeline runs daily, fetching all assets in ~54 seconds with automatic git commit/push. No blocking issues remain.

Bottom Line: Project operational in 2 days; synthetic scoring fully replaced with real computed data; automated daily pipeline active; 62 curated intel briefs deployed.

2. Objectives Review

Primary Objective Status

Objective	Success Criterion	Status	Confidence	Notes
Replace synthetic scoring with real data	All 5 dimensions use computed or fetched values	Complete	High	29 metrics across 5 dimensions
Compute technical indicators client-side	RSI, MACD, Bollinger, SMA, Volume from price history	Complete	High	6 indicators computed from 5-year history
Provide per-ticker intel briefs	Thesis/recent/risk/watch for all assets	Complete	High	62 briefs curated; externalized to JSON
Automate daily data pipeline	Scheduled fetch + git push	Complete	High	54 stocks + 8 crypto in ~54 seconds
Support portfolio categorization	Stakes, Watchlist, Favorites	Complete	High	Search, sort, collapsible sections

Objective	Success Criterion	Status	Confidence	Notes
Build responsive dashboard	Usable on desktop and tablet	Complete	High	Compact mode for < 768px width

Objective Health Assessment

On Track: - All scoring dimensions populated with real data - Pipeline runs successfully (54/54 stocks, 8/8 crypto on last run) - Intel briefs loaded for all 62 assets - Auto-refresh active (5 min prices, 4 hour intel)

At Risk: None

Blocked: None

3. Progress This Period

Work Completed

Item	Description	Outcome	Implications
Project scaffolding	React + Vite + Recharts setup with dark theme	Build pipeline functional	Clean development environment
Asset registry	54 stocks + 8 crypto defined with metadata	Full coverage of target portfolio	Covers 11 sectors + crypto
Data pipeline	Python script (yfinance + CoinGecko)	54 stocks + 8 crypto fetched in ~54s	Real-time data available
Synthetic scoring system	Initial RNG-based scoring	Placeholder functional	Later replaced entirely
Split-panel layout	Sidebar ticker list + detail panel	Core UI operational	Responsive with compact mode
Real data scoring	5 dimensions, 29 metrics from real sources	Synthetic fully replaced	Scores now meaningful and auditable
Technical indicators	RSI, MACD, Bollinger, SMA, Volume, Momentum	Computed client-side from history	No server-side computation needed
Normalization layer	0-100 scoring with inversion support	All metrics comparable	Raw values shown alongside scores
Intel briefs	62 per-ticker briefs with thesis/catalyst/risk/watch	Analyst-grade context for all assets	Externalized to JSON for easy editing
Score tooltip	Hover on ScoreRing shows intel + dimension bars	Quick intel access in overview tab	Visual grade + breakdown
Intel in dropdown	Expanded ticker card shows full intel inline	No hover required; always accessible	Better UX for detailed review
Automation	Daily pipeline + git push + log cleanup	Hands-off daily updates	7-day log retention
Price history	5-year daily history with range selector	Long-term trend analysis	1W/1M/3M/ALL selectable

Work in Progress

Item	Current State	% Complete	Expected Completion	Blockers
Intel brief automation	Manual curation only	0%	TBD	Need LLM API or news pipeline
Extended crypto history	CoinGecko free tier limited to 7-day sparkline	50%	TBD	Requires API key or alternative source

Work Not Started (Future)

Item	Priority	Reason Deferred
Score change alerts	Medium	Core dashboard needed first
On-chain crypto metrics	Low	Not available from current APIs
Multi-user support	Low	Personal dashboard; no need

4. Technical Details

Measurements and Data

Parameter	Value	Method	Date	Notes
Total assets monitored	62	Asset registry count	2026-02-06	54 stocks + 8 crypto
Pipeline execution time	54.1 seconds	last_update_status.json	2026-02-07	Including 54 stocks + 8 crypto
Stocks successfully fetched	54/54	Pipeline status	2026-02-07	0 failures
Crypto successfully fetched	8/8	Pipeline status	2026-02-07	All coins fetched
App.jsx lines of code	2,385	wc -l	2026-02-06	Single-file React app
Pipeline script lines	514	wc -l	2026-02-06	generate_market_data.py
Built JS bundle size	645.8 KB	Vite build output	2026-02-06	183 KB gzipped
Total git commits	34	git log count	2026-02-06	Over 2-day sprint
Metrics per stock	29	Dimension metric count	2026-02-06	6+6+6+6+5
Metrics per crypto	12	Technicals + Momentum only	2026-02-06	Other dimensions N/A
Intel briefs	62	intel_briefs.json	2026-02-06	All assets covered
Price history depth (stocks)	5 years	yfinance period="5y"	2026-02-06	~1,250 daily data points
Price history depth (crypto)	7 days	CoinGecko sparkline	2026-02-06	Free tier limitation

Scoring Validation

Dimension	Metric Count	Data Source	Computation	Validated
Fundamentals	6	yfinance stock.info	Normalize to 0-100	Yes
Technicals	6	Price history (client-side)	RSI/MACD/Bollinger/SMA/Vol/Trend	Yes
Valuation	6	yfinance stock.info	Normalize with inversion	Yes
Momentum	6	Price history (client-side)	Period returns + trend consistency	Yes
Analyst	5	yfinance stock.info	Recommendation mapping + normalize	Yes

Test Results

Test	Purpose	Result	Pass/Fail	Notes
Pipeline execution	All assets fetch without error	54 stocks, 8 crypto, 0 failures	Pass	~54 seconds execution
Vite build	App compiles without errors	Clean build, 645.8 KB bundle	Pass	Warning: chunk > 500KB
Data integrity	No NaN/Inf in JSON output	sanitize_for_json removes invalid values	Pass	Edge cases handled
Score computation	Scores vary meaningfully, not clustered	Range observed across assets	Pass	Not synthetic 43-60 clustering
Crypto dimension handling	Crypto shows only Technicals + Momentum	Other dims show N/A / hidden tabs	Pass	Correct behavior
Synthetic fallback	Missing tickers get generated data	Fallback to generateAssetData()	Pass	Marked with _synthetic flag
Auto-refresh	Prices refresh every 5 minutes	RefreshKey increments on interval	Pass	Intel refreshes every 4 hours
Responsive layout	Compact mode at < 768px	Layout adapts correctly	Pass	Split-panel collapses to single

Anomalies and Unexpected Observations

Observation	Expected	Actual	Explanation	Follow-up
CoinGecko market_chart returns 401	Extended crypto history	Free tier blocks this endpoint	Requires API key	Evaluate paid tier
Dropbox EPERM on build	Clean build	Permission denied on dist/index.html	Dropbox file lock	Retry resolves it

Observation	Expected	Actual	Explanation	Follow-up
Vite timestamp files accumulating CRWV (CoreWeave) limited data	Clean project root Full fundamentals	15+ .mjs temp files Sparse — recently IPO'd	Vite HMR artifacts yfinance coverage lag	Add to .gitignore Will improve over time

5. Issues and Risks

Active Issues

ID	Issue	Severity	Impact	Status	Resolution Plan
I-001	Intel briefs require manual curation	Medium	Briefs become stale within weeks	Open	Evaluate LLM API for semi-automated generation
I-002	Crypto history limited to 7-day sparkline	Low	Short-term analysis only for crypto	Open	Evaluate CoinGecko paid tier (\$130/mo) or alternatives
I-003	Bundle size exceeds 500KB warning	Low	Slightly slower initial load	Open	Consider code splitting with dynamic imports
I-004	Vite temp files in project root	Cosmetic	Cluttered project directory	Open	Add *.timestamp-*.mjs to .gitignore

Risk Register

ID	Risk	Probability	Impact	Mitigation	Contingency
R-001	yfinance API blocked or rate-limited	Low	No stock data; scores stale	Batch download reduces calls; individual fallback	Synthetic fallback; cached data still available
R-002	CoinGecko free tier deprecated	Medium	No crypto data	Minimal dependency (sparkline only)	Alternative: CoinMarketCap, CryptoCompare
R-003	Intel briefs become factually outdated	High	Misleading analyst context	Manual review schedule	Add "last updated" timestamps per brief

ID	Risk	Probability	Impact	Mitigation	Contingency
R-004	Normalization ranges miscalibrated	Medium	Scores meaningless for outliers	Audit extreme values; adjust ranges	Default to 50 (neutral) for out-of-range values

Structural Hurdles

Hurdle	Nature	Impact	What Would Help
Manual intel curation doesn't scale	Operational	62 briefs need periodic updates	LLM API integration; news aggregation pipeline
Single-file React app (2,385 lines)	Technical	Hard to maintain long-term	Component extraction into separate files
No backtesting of scoring model	Analytical	Unknown predictive value	Historical score computation; return correlation study

6. Critical Path and Dependencies

Critical Path Items

Item	Current Status	Required Completion	Risk Level
Daily pipeline operational	Complete	Complete (Feb 6)	Low
All dimensions scoring real data	Complete	Complete (Feb 6)	Low
Intel briefs for all assets	Complete	Complete (Feb 6)	Low
Intel brief refresh pipeline	Not Started	TBD	Medium
Scoring model validation	Not Started	Q2 2026	Medium

Dependencies

Dependency	Type	Source	Status	Impact if Disrupted
Yahoo Finance (yfinance)	External API	Yahoo	Active	No stock data; entire scoring system fails
CoinGecko	External API	CoinGecko	Active	No crypto data; 8 assets fall to synthetic
GitHub	External Service	GitHub	Active	Auto-push fails; data still local
Windows Task Scheduler	System	Local	Active	Manual runs required
Node.js / Vite	Build Tool	Local	Active	Cannot rebuild frontend

7. Resource Status

Personnel

Role	Allocation	Notes
Developer (Cole Prather)	Active	Full build + maintenance

Technology Stack

Resource	Status	Notes
React 18 + Vite 5	Active	Frontend build system
Python 3.8+	Active	Data pipeline
yfinance	Active	Stock data
CoinGecko (free tier)	Active	Crypto data
Git + GitHub	Active	Version control + hosting

Budget

Category	Cost	Notes
Data APIs	\$0	All free tier
Hosting	\$0	GitHub Pages via existing site
Dependencies	\$0	Open-source packages

8. Plan Forward

Immediate Priorities (Next 2-4 Weeks)

Priority	Action	Target Date	Success Criterion
1	Monitor pipeline reliability over 2 weeks	Feb 20	> 95% success rate
2	Audit scoring edge cases (negative margins, extreme P/E)	Feb 14	No anomalous scores
3	Add intel brief timestamps ("as of" dates)	Feb 14	Staleness visible to user
4	Clean up Vite temp files; update .gitignore	Feb 10	Clean project root

Upcoming Experiments

Experiment	Purpose	Target Date	Resources
Score-return correlation	Test if high composite scores predict 1M returns	Apr 2026	2 months of daily scores + returns
LLM intel generation	Test Claude API for automated brief writing	Mar 2026	API access + prompt engineering
CoinGecko paid tier eval	Assess extended crypto history value	Mar 2026	\$130/month budget decision

Milestones

Milestone	Target Date	Status	Notes
Initial build complete	Feb 5, 2026	Complete	Dashboard functional with synthetic data
Real scoring deployed	Feb 6, 2026	Complete	All 5 dimensions with real data
Intel briefs deployed	Feb 6, 2026	Complete	62 curated briefs
2-week reliability validated	Feb 20, 2026	In Progress	Pipeline monitoring
Scoring model backtested	Apr 2026	Planned	Need 2 months of score history
Intel automation prototype	Mar 2026	Planned	LLM API evaluation

9. Lessons and Observations

What's Working

- **yfinance batch download:** Fetching 54 stocks in one batch call is fast and reliable (~54 seconds total including crypto)
- **Client-side technical indicators:** Computing RSI/MACD/Bollinger from price history eliminates server-side computation dependency
- **Dual display (score + raw value):** Users see both the normalized score bar and the actual P/E ratio, margin percentage, etc.
- **Externalized intel briefs:** Editing JSON doesn't require a rebuild; 4-hour auto-refresh picks up changes
- **Synthetic fallback:** When live data is unavailable, assets still render with neutral scores and clear "synthetic" marker
- **Git auto-push:** Pipeline commits and pushes data updates automatically — zero manual intervention for daily updates

What's Not Working

- **Single-file architecture:** 2,385 lines in App.jsx is workable but will become unwieldy with further features
- **Manual intel curation:** Writing 62 analyst briefs is time-intensive and they go stale quickly
- **Crypto data depth:** 7-day sparkline is insufficient for meaningful technical analysis on crypto assets

- **No score history:** Current scores are point-in-time; no tracking of how scores change over time

Insights Gained

- **Normalization range selection matters enormously:** A P/E range of 0-60 (inverted) produces very different distributions than 0-100. Ranges were tuned empirically against the 54-stock universe.
- **RSI scoring needs special treatment:** Linear normalization doesn't work for RSI because both extremes (overbought >70, oversold <30) are bearish signals. Distance-from-50 scoring captures this correctly.
- **Intel briefs transform the experience:** Moving from generic grade descriptions ("Top-tier across most dimensions") to specific analyst context ("NVDA: Blackwell ramp fastest in history, 70% of Q4 data center revenue") made the dashboard dramatically more useful.
- **Crypto needs different treatment:** Forcing stock-like dimensions onto crypto assets produces meaningless scores. The decision to show only Technicals + Momentum for crypto was correct.

Recommendations

1. **Track score history:** Store daily composite scores to enable trend analysis and backtesting
2. **Component extraction:** Split App.jsx into TickerCard.jsx, DetailPanel.jsx, ScoreTooltip.jsx, etc.
3. **Intel brief versioning:** Add "lastUpdated" per brief; flag briefs older than 30 days
4. **Normalization audit:** Review and document the rationale for each metric's min/max/inversion
5. **CoinGecko upgrade evaluation:** \$130/month for Pro tier may be justified by extended crypto history

10. Open Questions and Uncertainties

Unresolved Questions

Question	Why It Matters	What Would Answer It	Priority
Do composite scores correlate with forward returns?	Validates the entire scoring model	3-6 months of score history + return analysis	High
Are normalization ranges optimal?	Extreme values may skew scores	Statistical analysis of score distributions across all tickers	Medium
Should intel briefs be LLM-generated?	Manual curation doesn't scale	Prototype with Claude API; compare quality	Medium
Is the 5-dimension model the right decomposition?	Could be over-fitting or under-fitting	Compare with established models (Piotroski F-Score, etc.)	Low

Assumptions Requiring Validation

Assumption	Current Confidence	How to Validate
Higher composite scores indicate better investments	Low (untested)	Backtest against 3-6 month returns

Assumption	Current Confidence	How to Validate
RSI distance-from-50 scoring is correct	Medium	Compare with traditional overbought/oversold interpretation
yfinance data is accurate and timely	High	Spot-check against official exchange data
Intel briefs provide actionable context	Medium (subjective)	User feedback over 1 month of use

Appendix A: Asset Universe

Stakes (35 positions)

AVGO, AMD, MU, MCHP, NVDA, INTC, ASML, PLTR, SMC, CRWV, AMZN, QBTS, RGTI, RKL, KTOS, ASTS, LMT, GE, TSLA, QS, ARRY, SPOT, CHWY, DIS, AMC, DJT, LULU, AEO, GME, CBRL, CAVA, TSCO, MPW, COMP, LEG

Watchlist (9 tickers)

AAPL, MSFT, GOOGL, META, IONQ, COIN, BTC, ETH, SOL

Additional Coverage (18 tickers)

TSM, ASML, ENPH, SQ, HOOD, SOFI, MRNA, CRSP, NET, SNOW, DDOG, RIVN, LCID, LUNR, RDW, XRP, ADA, DOGE, AVAX, LINK

Appendix B: Change Log

Version	Date	Author	Changes
1.0	2026-02-06	Cole Prather	Initial PSR — project operational with full scoring

This status report covers the initial build sprint. Future reports will focus on pipeline reliability, scoring validation, and intel brief maintenance.